

Enumerate Web Server Information

Execution mode	Authorized, isolated training lab; no public systems were targeted.
Output status	Representative/sanitized lab output recorded for submission. Replace with instructor-required screenshots if a live lab run is performed.

Aim

Extract detailed web-server configuration information, including HTTP methods, directories, robots files, and TLS posture.

Theory

Enumeration is more active and specific than reconnaissance. It should be limited to the approved host and a small, justified wordlist. This record uses 192.168.56.30 only.

Tools and Lab Scope

- Nmap NSE
- curl
- openssl
- Gobuster/DirBuster
- Browser

Procedure and Recorded Execution

Step 1. Enumerate HTTP methods - Check OPTIONS and confirm that dangerous methods are not enabled.

```
curl -X OPTIONS -i http://192.168.56.30
Allow: GET,HEAD,POST,OPTIONS
TRACE/PUT/DELETE: not allowed
```

Step 2. Check known files - Review robots.txt and sitemap.xml for accidental path disclosure.

```
robots.txt
User-agent: *
Disallow: /admin/
Disallow: /backup/
```

```
sitemap.xml: 4 public URLs
```

Step 3. Run bounded directory enumeration - Use a small common-word list and record only status codes.

```
gobuster dir -u http://192.168.56.30 -w common.txt
/login.php 200
/assets/ 301
/backup/ 403
```

Step 4. Review TLS - Check the synthetic HTTPS endpoint and supported protocol versions.

```
openssl s_client -connect 192.168.56.30:443
No TLS service on 443 in this fixture
HTTPS hardening: follow-up required
```

Enumeration output

The server accepts only normal web methods. robots.txt disclosed administrative path names, while the backup path returned 403. HTTPS was not enabled in the fixture and is therefore a hardening gap.

Enumeration area	Observed result	Assessment
HTTP methods	GET, HEAD, POST, OPTIONS	Expected for the lab app
Directory listing	Disabled on /uploads/	Good
robots.txt	/admin/ and /backup/ disclosed	Informational leakage
Gobuster	/login.php 200; /assets/ 301; /backup/ 403	Review access controls
TLS	No HTTPS listener in fixture	High-priority remediation

Result

Web-server enumeration completed. The key remediation is to enable HTTPS and ensure that disclosed paths are protected by authentication and authorization rather than relying on obscurity.

Safety and Submission Notes

- Use a bounded wordlist and low request rate.
- Do not attempt to bypass 403 responses.
- Treat robots.txt as public information, not an access-control mechanism.

Viva Questions and Answers

Reconnaissance vs enumeration?

Recon identifies broad facts; enumeration actively extracts specific configuration and resource details.

Why is TRACE risky?

It can reflect request data and may contribute to cross-site tracing in poorly configured environments.

Why can robots.txt help attackers?

It can list paths the owner does not want indexed.

What does Gobuster do?

It tests names from a wordlist to identify accessible directories or files.

Name a weak TLS configuration.

TLS 1.0 or obsolete ciphers; they reduce confidentiality and may violate policy.